

SIMATS ENGINEERING ASSESSMENT TOOL

COURSE NAME: Theory of Computation

COURSE CODE: CSA13

COURSE OUTCOME COVERED

CO2: Design Finite Automata DFA, NFA, ϵ -NFA and demonstrate their equivalence for recognizing regular languages. (BL:3)

Assessment Tool Weightage: 10%

QUESTIONS: 5
Duration : 1 Hour

Total Marks:25

Application Based Question

Code of Conduct:

I V. Smylka (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Smylka
Signature of the student:

DFA for passwords ending with 01

Given $\Sigma = \{0, 1\}$, construct a DFA that accepts strings ending in 01.

States:

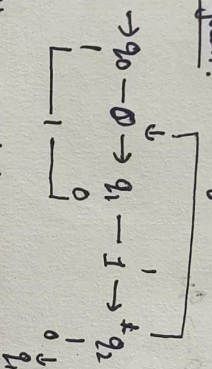
- q_0 - Initial state
- q_1 - Last symbol is 0
- q_2 - String ends in 01

Transition Table:

State	0	1
$\rightarrow q_0$	q_1	q_0
q_1	q_1	q_2
q_2	q_1	q_0

Here, q_0 is the initial state and q_2 is the final / accepting state.

State Diagram:



Ex:

- 1101 $\rightarrow$ Accepted
- 1001 $\rightarrow$ Accepted
- 01 $\rightarrow$ Accepted
- 101 $\rightarrow$ Accepted
- 1100 $\rightarrow$ Rejected
- 1110 $\rightarrow$ Rejected

The DFA has 3 states and is already minimized because all states have different roles.

Efficiency: A minimized DFA requires less hardware, memory, and power and provides fast password validation.

Conclusion: The 3-state DFA is suitable for embedded security and authentication systems.

2. Regular Expression for aaaaabb

The required regular expression is: aaaaabb

It accepts only the string aaaaabb.

Regular expressions are widely used in web applications for:

- Email validation
- Password validation
- Phone numbers
- Usernames
- Form inputs

Equivalent Finite Automaton

$\rightarrow q_0 \xrightarrow{a} q_1 \xrightarrow{a} q_2 \xrightarrow{a} q_3$

$q_3 \xrightarrow{a} q_4$

|

b

↓

q_5

|

b

↓

* q_6

Ex:

<u>Input</u>	<u>Result</u>
aaaaabb	Accepted
aaabb	Rejected
aaaab	Rejected
aaaabbb	Rejected
aabbbb	Rejected

Advantages:

- Simple pattern specification
- Fast validation
- Reduced programming effort
- Consistent input checking
- Improved data correctness
- Easy rejection of invalid input

conversion of a^+b^+ into an equivalent Finite Automaton

Pattern recognition is important in intrusion detection systems, compiler and real-time monitoring applications. Regular expressions provide a convenient way to specify patterns, while finite automata can recognize those patterns efficiently.

Given: a^+b^+

Meaning: a^+ means 2 or more a 's.
 b^+ means 2 or more b 's.

Ex:

ϵ
 a
 aaa
 b
 bdb
 $caabbb$

FA construction

States:

- q_0 - Reading a 's
- q_1 - Reading b 's
- q_d - Dead state

Transition Table

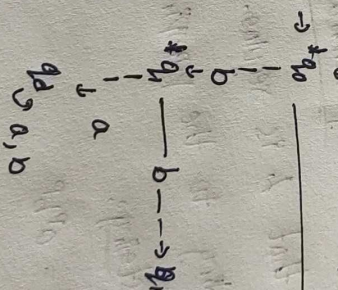
State	a	b
$\rightarrow q_0$	q_0	q_1
$+q_1$	q_d	q_1
q_d	q_d	q_d

Real-Time Applications

Finite automata are useful for:

- Intrusion detection
- Network traffic monitoring
- Spam filtering
- Compiler lexical analysis
- Log monitoring
- Input validation

State Diagram



4. Pumping Lemma ~~Proof~~

Finite automata have limited memory and cannot recognize languages that require unlimited counting or comparison. The Pumping Lemma can be used to prove that the language

$$L = \{a^n b^n \mid n \geq 0\}$$

is not regular.

Given Language

$$L = \{a^n b^n \mid n \geq 0\}$$

Ex:

ϵ

ab

aabb

aaabbb

aaaabbbb

Every string must contain an equal number of as and bs.

Proof Using Pumping Lemma

Assume that L is regular.

According to the pumping lemma, there exists a pumping length

Choose the string:

$$w = a^p b^p$$

Conclusion:

The pumping lemma proves that $\{a^n b^n \mid n \geq 0\}$ is not regular.

This demonstrates why finite automata alone are insufficient for all complex tasks and why more powerful computational models are required for syntax analysis.

Closure Properties of Regular Languages

Modern applications such as intrusion detection, spam filtering and input validation often require several pattern-matching rules to be combined. Closure properties of regular languages allow rules to be combined while preserving regularity.

Closure Property

A language family is said to be closed under an operation if applying that operation to languages in the family produces another language in the same family.

Regular languages are closed under several operations.

Important closure properties

If L_1 and L_2 are regular languages, then:

1. Union:

$$L_1 \cup L_2$$

The resulting language is regular.

It represents strings belonging to either L_1 or L_2 .

2. Intersection:

$$L_1 \cap L_2$$

The resulting language is regular.

It represents strings belonging to both languages.

3. Complement:

$$\bar{L}$$

The complement of a regular language is also regular.

4. Difference

$L_1 - L_2$

The resulting language is a regular.

5. Concatenation

$L_1 L_2$

The resulting language is regular.

6. Kleene star

L_1^*

The resulting language is regular.

Advantages

Using closure properties provides:

1. Easy combination of multiple rules.
2. Efficient implementation using finite automata.
3. Fast pattern matching.
4. Predictable processing time.
5. Low memory requirements.
6. Suitability for real-time applications.

SIMATS ENGINEERING ASSESSMENT TOOL

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

92/100

[Signature]